

From Remote Space to Remote Time

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Abstract

We explore a design vision that reimagines human connection not only across distance but also across time, extending even beyond physical lifespans. While Telepresence was developed to connect remote spaces, this panel shifts the focus to remote time—the desire to be remembered, to reconnect with the past, and to speculate on possible futures. We bring together perspectives from generative AI, cultural heritage, and HCI to examine how connection across time can be mediated and reinterpreted. AI researchers develop synthetic agents "ghost bots" that reconstruct identities from fragmented data. Scholars in archaeological heritage investigate how artifacts carry memory across centuries, offering insight into how cultural traces become embodied knowledge. HCI researchers study interactive systems that honor the presence of absence. Together, this interdisciplinary dialogue across digital reconstruction, archaeological preservation, and practices of cherishing material traces raises broader philosophical questions about how emerging technologies shape relationships across past, present, and future.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; *HCI theory, concepts and models*.

Keywords

telepresence, teleabsence, synthetic presence, cultural heritage, material traces, memory, remembrance, grief, afterlife, wellbeing

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1 Introduction and Motivation

Research on interaction across space—particularly telepresence that enables people to collaborate, communicate, and share presence despite physical distance—has been widely explored in HCI. [6, 9] As advances in data collection and preservation, machine learning, generative models, and multimodal media continue, they have sparked growing interest in forms of interaction that unfold not only across space but also across time—engaging with memories, and traces that extend beyond the immediate present and connect the past, present, and future. [1, 2]

Many of our deepest longings are not only spatial but also temporal: the desire to remember, to be remembered, to revisit moments that have passed, or to reach toward futures we will never see. Across disciplines, researchers have explored these temporal desires by treating memories and traces not as static records, but as active, evolving material that can be interpreted, cherished, or re-experienced in new ways.

Archaeological preservation practices demonstrate that the material past—objects that often outlast human memory and embed traces of lived experience—is always encountered through incomplete and time-worn fragments, each shaped by decay, environmental conditions, and the partial nature of what survives. [3, 4]



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Rather than attempting to reconstruct a complete or literal whole, researchers with archaeological insight focus on interpreting these remnants—reading traces, respecting gaps, and acknowledging what cannot be known.

At the other temporal extreme, AI researchers investigate digital afterlives, online memorialization, and the management of identity data after death. Work on synthetic humans and “ghost bots” [10, 11] seeks to reconstruct, simulate, or extend aspects of human presence through generative AI. These systems illustrate both promise and tension: while they may offer companionship or emotional support, they also risk producing PseudoPresence—generating novel utterances that may distort or overwrite the authentic traces of an individual. They may further introduce Prosthetic Memories—“second-hand” memories acquired through media rather than direct experience—which can still feel personal and emotionally real, complicating how we remember. Although such interactions may bring comfort or familiarity, they can reshape a person’s memory through invention rather than preservation.

Complementing these trajectories, some works in wellbeing demonstrate how subtle material artifacts can support remembrance and emotional continuity without attempting to replicate the past. Research on ongoingness suggests a dialogical relationship between the living and the dead, in which data related to a deceased person can be remediated into an active, evolving exchange. This is not remembrance or memorialization, but an understanding that both the living and the dead continue to “become,” and that their data offers us ways in to what they might say now, rather than what they said when alive. This approach emphasizes care, interpretation, and the value of authentic absence, offering a counterbalance to generative or synthetic forms of reanimation. [13, 14]

In a similar spirit, the vision of TeleAbsence [7] highlights the importance of honoring the authenticity of absence, showing that subtle, trace-based engagements can evoke presence through the genuine remnants people leave behind. [8, 12] Drawing on the emotional notion of “saudade” —cherishing the emptiness that remains—TeleAbsence resists filling absence with synthetic invention, instead preserving ambiguity and emotional depth under the belief that “what fades is not necessarily lost,” captured in the metaphor: “an amber remembers the light it once held; an ember remembers the warmth it once gave.” [5] This approach invites engagement with silence, incompleteness, and resonance, as seen in works such as *MirrorFugue* [15], which layers past piano performances onto present bodies; *TwitterPoet*, which remixes existing traces into reflective interactions; and cultural practices like the *Wind Phone*, a disconnected telephone used to speak symbolically to the deceased without synthetic substitution [7].

Together, these perspectives illuminate how technologies shape what is remembered, what is forgotten, and how emotional connection persists across time, revealing diverse ways to engage with loss, grief, legacy, and the enduring desire to remain connected across temporal distance.

2 Organizers, Moderators and Panelists

The panel features two organizers, one moderator, and five invited panelists, and brings together HCI researchers, HAI (Human–AI Interaction) researchers, Telepresence/VR researchers, and designers

whose work redefines how people connect with memory, absence, and presence across time. The group represents diversity across geographic regions, including Asian countries such as Korea, Japan, Thailand, and China, and European countries such as France, the Netherlands, and the UK, bringing together perspectives shaped by different cultural understandings of memory, ancestry, and temporal connection.

2.1 Co-Organizers

Prof. Hiroshi Ishii is Jerome B. Wiesner Professor of Media Arts and Sciences, Associate Director of the MIT Media Lab, and Director of the Tangible Media Group. He is a pioneering figure in Human–Computer Interaction whose research has transformed how people relate to digital information through the visions of Tangible Bits, which transform digital information into tangible form, and Radical Atoms, which imagine future materials that can change their physical properties in real time across space. He now focuses on a new vision TeleAbsence—enabling interactions with the presence of absence across time. More deeply, TeleAbsence explores connections with past selves, ancestors, or future generations, addressing universal longings such as “I want to always remember you” and “I want to always be remembered by you.” In this panel, he provides the philosophical foundation for designing across temporal boundaries, creating new forms of presence that honor absence, preserve meaning, and extend human connection beyond a single lifespan.

Hye Jun Youn is a graduate researcher at the MIT Media Lab whose work explores human–material interfaces within the vision of Radical Atoms and extends these ideas into the temporal domain through the emerging concept of TeleAbsence. As a co-organizer of this panel, Hye Jun focuses on shaping the structure of the session, designing interactive prompts or interfaces that actively involve attendees, and fostering meaningful audience engagement.

2.2 Moderator (MC)

Dr. Jie Li is a HCI researcher and creative designer specializing in immersive experiences and human–AI symbiosis. Her work investigates how interactive systems can shape perception, emotional experience, and the evolving relationships between humans and intelligent agents. She recently contributed the article “Eternal Life?” to *Interactions* (Volume 32, Issue 3), exploring how AI-mediated presence challenges traditional notions of identity, and memory. Dr. Li will guide the session’s flow, introduce core themes, and balance contrasting perspectives. She will also probe how different disciplines imagine connection across time—challenging familiar notions of remembering and presence. By placing archaeological views on preservation, HCI approaches to embodied interaction, and AI-engineering visions of reconstructed presence in conversation, she will highlight both tensions and shared insights.

2.3 Panelist

Dr. Xiao Xiao is a researcher, designer, and musician, and is Director and Principal Investigator of the Institute for Future Technologies at Da Vinci Higher Education in Paris. She was previously a Research Affiliate in the Tangible Media Group at the MIT Media Lab. Her work examines how technology mediates emotional presence

across time. She explores illusory communication, embodied memory, and generative AI as a co-performer rather than a tool, offering a critical perspective on the future of human–machine relationships. Through projects such as *MirrorFugue*, which enables duets with pianists from the past, she presents one of the earliest empirical studies of temporal interaction through performance—revealing how sound, gesture, and expression can reconnect us with those who are absent, unreachable, or no longer alive. Dr. Xiao brings a perspective grounded in artistic inquiry and embodied performance, highlighting how temporal connection can be designed not through replication, but through expressive and improvisational engagement with traces of the past.

Prof. Eugene Ch'ng is Dean of the School of Culture and Creativity at Beijing Normal–Hong Kong Baptist University and Director of the Centre for Computational Culture and Heritage | NVIDIA DLI. A leading scholar in digital heritage and XR, he is Editor-in-Chief of *PRESENCE: Virtual and Augmented Reality* (MIT Press), and his work on data-driven, immersive approaches to cultural memory has resulted in over 150 publications and major international research funding. His research examines how virtual reconstructions, XR environments, and large-scale cultural datasets can revive and reinterpret historical sites, rituals, and collective memory for contemporary audiences. Through projects that simulate ancient environments, visualize archaeological data, and reconstruct lost practices, he investigates how technology can reconnect people with distant or fragmented pasts. His perspective offers an archaeological and cultural heritage lens within the panel, highlighting how temporal connection is shaped through interpretation, reconstruction, and access to material traces.

Associate Prof. Jed Brubaker is a leading researcher in digital identity, social computing, and technology ethics at the University of Colorado Boulder. He is known for his groundbreaking work on digital afterlives, online memorialization, and the management of identity data after death. As an expert on post-mortem interactions and digital legacy, his research examines how social technologies shape grieving, remembrance, and ongoing relationships with those who have passed, as well as how platforms mediate these connections through profiles, archives, and algorithmic systems. His work reveals the complexities of identity continuity—who controls data after death, how digital traces persist or evolve, and how emotional and cultural practices intersect with technological infrastructures. In this panel, Dr. Brubaker brings a critical, human-centered perspective on the ethical and psychological challenges of technologies that evoke the presence of deceased loved ones, offering insight into distinguishing healthy remembrance from potentially harmful synthetic representations and illuminating ethical, emotional, and sociotechnical dimensions of connecting across time.

Prof. Jayne Wallace is Professor of Craft, Design, and Well-being at Northumbria University, known for pioneering work at the intersection of craft, HCI, and care. Her practice investigates how material artifacts—such as digital jewelry, objects of ongoingness, and tools for dementia care—support remembrance and emotional continuity without reproducing the past literally, emphasizing how subtle traces can honor loss and sustain relationships across time. Prof. Wallace brings a grounded approach to designing intimate, meaningful interactions that preserve personal histories and support enduring connections across temporal distance.

Assistant Prof. Pat Pataranutaporn is a scientist, designer, and media researcher at the MIT Media Lab whose work explores the future of human–AI interaction and synthetic media. Through projects involving AI-driven avatars, synthetic humans, and experimental future interfaces, he investigates how computational systems can reconstruct, simulate, or extend aspects of human presence beyond traditional physical or temporal limits. His research into AI-generated characters for well-being offers a forward-looking view of how AI-mediated identities may reshape memory, representation, and the continuity of self. He introduces the rapidly evolving domain of *PseudoPresence*, where AI systems create highly realistic synthetic representations of people. His viewpoint raises essential questions about authenticity, illusion, and the ethical boundaries of designing presence across time.

3 Discussion Format

The discussion will fold into three topics. Topic 1 focuses on transitions from presence to absence, as panelists discuss how their work engages communication across space and time. Topic 2 examines archiving and curating the past, bringing perspectives on XR reconstructions, AI-generated digital traces, and musical or cultural data collection. Topic 3 explores the tension between *Synthetic Presence* and *TeleAbsence*, comparing AI-generated identities with physical mementos, material artifacts, and embodied performance.

3.1 Across Space to Across Time: Presence to Absence

The discussion begins with technologies that connect people across space—through VR environments, ambient media, remote performance, and tangible artifacts. These examples illustrate how spatial connection forms a foundation for deeper questions about connection across time. Rather than linking only those who are physically distant, such technologies open possibilities for conversing with past versions of ourselves, reinterpreting the traces of loved ones who have passed, or sending messages and intentions to imagined future generations.

Some foundational questions will guide the conversation: How does designing communication across time differ from designing communication across distance? What types of emotional connection emerge when we interact with someone from another moment—whether a past self, a loved one who is gone, or future people we will never meet? What memory, message, or trace would you choose to hold onto, send forward, or protect across time? What would your “*TeleAbsence* moment” be? These prompts also invite participants to reflect on their personal relationship with memory, loss, and continuity, exploring how presence can persist beyond the limits of the present moment.

3.2 Archiving and Curating the Past

The conversation now shifts toward how the past is collected, interpreted, and preserved. The moderator begins by raising questions about representation and decision-making: What level of clarity or accuracy do we expect from the past? How do we collect, select, and privilege certain data as “the past,” and who has the authority to make these choices? What are the methods for capturing and organizing information—lifelong wearable devices that gather

daily traces, or automated data collection across digital platforms? Should the past be accurately replicated, or is ambiguity essential to how it is remembered? These questions reveal how different disciplines decide what to preserve, reinterpret, or let fade—showing how design choices shape the meaning and cultural value of the past.

In cultural heritage and archaeological work, reconstructing ancient sites do not restore a perfect past; they construct interpretive models grounded in evidence and open to revision. The past rarely survives intact—it arrives as fragments shaped by decay, loss, and time. Archaeologists therefore read both traces and absences as evidence, recognizing that ambiguity is not an error to eliminate but a fundamental condition of understanding history.

From an AI engineering standpoint, digital traces are not simply preserved—they are computationally extended. Generative systems infer missing details and produce new representations—voices, gestures, or identities that feel authentic despite never having existed in recorded form. Working from partial data, AI constructs coherent outputs from incomplete evidence, enabling new forms of posthumous presence while complicating cultural practices of mourning and digital legacy. Rather than maintaining memory as it was, generative AI creates memory that might have been, raising questions about authenticity, consent, and the boundary between preservation and fabrication.

Designers and HCI researchers craft interactive systems that recognize uncertainty and incompleteness as meaningful. Rather than reproducing what was lost, these practices engage with what remains—inviting reflection, imagination, and dialogue between presence and absence. They emphasize the authenticity of absence and silence, treating missing information as integral to how people remember, grieve, and stay connected. In this context, crafted mementos and digital memorials aim not to recreate the past too literally, but to honor what endures. Each discipline navigates its own balance between clarity and ambiguity, reconstruction and restraint, showing how the fragments preserved—and the absences left untouched—both shape how the past is carried into the future.

3.3 Synthetic Presence vs TeleAbsence

The discussion now shifts to designing connections beyond physical lifespans, raising questions of responsibility and choice: What should be preserved, reinterpreted, or allowed to fade? How much speculation is acceptable when reconstructing what cannot be known? Should technology help us live with loss rather than override it? Does generative AI fulfill longing or create illusory presence that obscures authentic grief? Where do we draw the line between cherishing traces and fabricating synthetic reconstruction?

With the rise of generative AI, the identities of the deceased are increasingly shaped through synthetic media and computational simulation. PseudoPresence (synthetic presence)—AI-driven “ghost bots” or “death bots” that generate new speech, behaviors, or responses on behalf of individuals who are no longer alive—raises questions about authenticity, consent, and the ethical boundaries of representing someone beyond their lifetime. These systems also raise concerns about how easily synthetic identities might overwrite real memories or construct false narratives about a person who can no longer speak for themselves.

A contrasting trajectory is offered by TeleAbsence, advocating authentic traces rather than synthetic presence. It does not attempt to speak for the deceased, but instead works with the fragments they genuinely leave behind—photographs, recordings (e.g., piano playing), handwriting (e.g., calligraphy, letters, postcards), performances, sounds, or objects—to create subtle, poetic, and reflective encounters to extend beyond physical lifespans. Viewed together, PseudoPresence (synthetic presence) and TeleAbsence reveal two contrasting visions for the future of posthumous interaction: one seeking to recreate and simulate, the other striving to preserve, interpret, and move forward.

4 Audience Engagement

This panel is an interactive, discussion-oriented forum that engages both expert panelists and the CHI audience in cross-disciplinary exploration. After brief introductory presentations, the session will unfold as a moderated debate and collaborative dialogue, encouraging diverse viewpoints and thought-provoking exchange. The audience will include a diverse mix of researchers, practitioners, and students in HCI, AI, XR, cultural heritage, and digital wellbeing. By bringing together communities that rarely intersect, the panel invites active participation—encouraging attendees to share reflections, respond to speculative prompts, and contribute personal rituals of remembrance.

4.1 Panel Structure and Engagement

The session will follow a structured 90-minute format led by a moderator. It begins with a 5-minute opening in which the moderator introduces the panel’s core theme and briefly presents each panelist, highlighting their background, research, and perspectives on engaging with the past, present, and future.

The introduction is followed by 20 minutes of lightning statements, during which each panelist briefly offers a spectrum of viewpoints on designing for connection beyond physical lifespans across HCI, Human–AI interaction, XR, and design for wellbeing. In addition to academic perspectives, panelists will share how they personally engage with memory, loss, and traces of the past.

During this phase, a live web interface will open, inviting panelists and audience members to share meaningful past moments—memories they wish to remember, memories they have forgotten, and moments they hope to pass forward. All contributions will be dynamically displayed on a large visual projection, forming a collective, evolving memory interface that will serve as a reference throughout the discussion, ensuring that audience voices actively shape the conversation.

The panel will then move through three thematic topics, each lasting 15 minutes. By juxtaposing how panelists work with past traces, the session brings generative AI, archaeological interpretation, HCI theory, and design practices that honor absence into direct dialogue. The moderator will facilitate the discussion by tying panelists’ perspectives to audience contributions, allowing tensions around authenticity, cultural diversity, ethical responsibility, and emotional mediation to surface through both alignment and disagreement.

The session concludes with a 20-minute audience engagement segment that incorporates live polling and speculative scenario

questions. Participants will reflect on prompts such as “Which would you choose: synthetic presence or TeleAbsence, and why?” and “How do you emotionally respond to different approaches to posthumous connection?” Polling will reveal areas of consensus, disagreement, and unexpected insight. The panelists will then offer closing provocations, highlighting future design challenges for the CHI community. This structure transforms the session from a one-directional presentation into a collaborative inquiry, inviting participants to shape how we might one day design connections across time.

From a logistical standpoint, the format requires only standard audiovisual equipment: a projector, microphones for panelists and audience members, and internet access for polling and interactive engagement. The structure ensures full accessibility and participation for attendees across modalities, creating an engaging, memorable, and thought-provoking experience well-suited to the interdisciplinary CHI community.

5 Conclusion

This panel invites the CHI community to examine how emerging technologies shape human connection across time. Interdisciplinary perspectives—including archaeological interpretation, synthetic presence, ongoingness, and the value of honoring absence—offer distinct and often conflicting ways of engaging with remembrance, mourning, legacy, and identity. By placing these perspectives in dialogue, rather than advancing a single vision, the panel encourages reflection on how contrasting strategies may coexist or diverge. In doing so, it contributes to the CHI community by identifying ethical, cultural, and methodological directions for future research on how technologies mediate mourning, legacy, and identity across time.

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